

	NOTA TECNICA TECHNICAL REPORT		Núm. No. ME-T-NT-110086
	Departamento Department	Methods	Avión Aircraft
Título/Title Program PANCO, Version 3			
Palabras clave/Key words MANUALS, BUCKLING, COMPUTER PROGRAMS			Clasificación acceso Access class P1 Registro de Revisiones RevisionRecord Pag. 2
Resumen/Summary This Technical Report describes the utilisation procedure of the PANCO Program, Version 3. The PANCO Program calculates reserve factors and applied and allowable stresses of uniform and non uniform metallic columns, with different support conditions under the action of axial and/or transversal loads.			
PROPIEDAD DE EADS CASA, Sociedad Unipersonal. Este documento no puede ser utilizado ni reproducido total o parcialmente sin la previa autorización escrita de la Dirección de Ingeniería y Tecnología de EADS-CASA. EADS CASA, Sociedad Unipersonal PROPERTY. This document shall neither be used nor completely or partially reproduced without previous written authorization by EADS-CASA Engineering and Technology Directorate.			
Distribución/Distribution	Realización/Execution		Fecha 1ª edición 1 st issue Date May 2011
Archivo de Cálculo	Realizado Prepared		Firma Signature
Jefe de la Subd. de Cálculo (*)	Álvaro García Herrero		
Jefes de Dpto. de Cálculo (*)			
Jefes de Sección de Cálculo			
	Comprobado Checked	Álvaro García Herrero	
(*) Cover sheet only	Aprobado Approved	Darío Fernández Villalba	

REGISTRO DE REVISIONES/REVISIONS RECORD

[illegible]

Table of Contents

1.- Technical sheet of the program	4
2.- Introduction.....	7
2.1 Program execution	10
3.- Input data file	11
4.- Output results.....	16
4.1 Output file	16
4.2 Error messages	17
5.- Windows application	19
6.- Examples	21
6.1 Example 1	21
6.2 Example 2	25

List of Figures

Figure 1: Coordinate system	7
Figure 2: Input data form.	20
Figure 3: Example 1	21
Figure 4: Example 2	25

1.- Technical sheet of the program

Technical Report: **ME-T-NT-110086**

PROGRAM

PANCO

REFERENCE

DATE

May 2011

OBJECTIVE

This program was developed to columns, with different support conditions under the action of axial and/or transversal loads.

LANGUAGE

Fortran 77

COMPUTER

Sun Solaris

NUMBER OF LINES (APROX.)

8000

EXECUTION MODE

Interactive

AUTHOR

J. C. Gómez

VERSION

1.0

Álvaro García Herrero

3.0

ADDITIONAL DOCUMENTS

Technical Report: Nota Técnica NT – 00 – ECE - 042
Manual Teórico del Programa PANCO, Versión 1.0

Technical Report: Nota Técnica NT – 00 – ECE - 043
Manual del Programador del Programa PANCO, Version 1.0

Notation

E_t	Tension Elastic Modulus
E_c	Compression Elastic Modulus
G	Shear Elastic Modulus
F_{tu}	Ultimate allowable stress
F_{ty}	Tension Yield allowable stress
F_{cy}	Compression Yield allowable stress
e	Ultimate Strain (in percentage)
L	Column length
A	Cross Section Area
I_{yy}	Cross Section Moment of Inertia about Y-axis
I_{zz}	Cross Section Moment of Inertia about Z-axis
J	Cross Section Torsion constant
R	Cross Section warping constant
y_0	Y coordinate of the shear center respect to the center of gravity
z_0	Z coordinate of the shear center respect to the center of gravity
F_{cc}	Crippling allowable stress
D_s	Distance from de Neutral Axis to the upper fiber (in the same plane that the transversal applied loads)
D_i	Distance from de Neutral Axis to the lower fiber (in the same plane that the transversal applied loads)
N_g	Number of seccions for geometry analysis along the column
x_i	X Coordinate of the section i
A_i	Cross section area at Section i
I_i	Moment of Inertia about the axis perpendicular to the applied bending loads plane at Section i
DS_i	Distance from de Neutral Axis to the upper fiber (in the same plane that the transversal applied loads) at Section i
DL_i	Distance from de Neutral Axis to the lower fiber (in the same plane that the transversal applied loads) at Section i
WP	Working plane
N_{px}	Number of Sections where a discrete axial load is applied
x_{pxi}	X Coordinate where the discrete axial load i is applied
F_{x1i}	Value of the discrete axial load i
N_{qx}	Number of Sections where a distributed axial load is applied

Notation (cont'd)

x_{qxi}	X Coordinate where the distributed axial load i is applied
q_{xi}	Value of the distributed axial load at section i
N_{pt}	Number of Sections where a discrete transverse load is applied
x_{pti}	X Coordinate where the discrete transverse load i is applied
P_t	Value of the discrete transvers load i_i
N_{qt}	Number of Sections where a distributed transversal load is applied
x_{qti}	X Coordinate where the distributed transversal load i is applied
q_{ti}	Value of the distributed transversal load i
N_M	Number of Sections where a discrete bending moment is applied
x_{Mi}	X Coordinate where the discrete bending moment i is applied
M_i	Value of the discrete bending moment i

2.- Introduction

The PANCO Program calculates reserve factors and applied and allowable stresses of uniform and non uniform metallic columns, with different support conditions under the action of axial and/or transversal loads.

A uniform column have a constant cross section along its complete length, and the applied axial load it's only applied at the ends of the beam.

Figure 1 below shows the coordinate system.

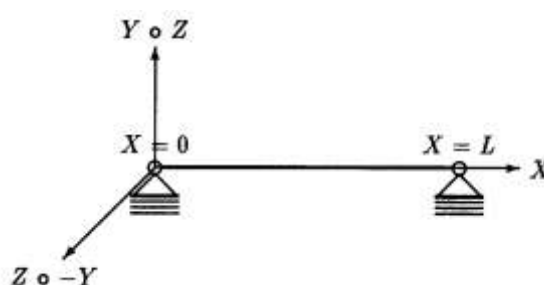


Figure 1:Coordinate system

A. The possible support conditions are the following ones:

Uniform Column:

a) Bending:

$X=0$

Simply supported
Clamped
Simply supported
Free
Simply supported
Supported
Supported

$X=L$

Simply supported
Clamped
Clamped
Clamped
Supported
Clamped
Supported

b) Torsion:

$X=0$

Free warping
Constrained warping
Free warping
Constrained warping
Gyration is always constrained at both ends

$X=L$

Free warping
Constrained warping
Constrained warping
Free warping

Non Uniform Column:

It is only analysed under bending loads, assuming the same support conditions as the uniform column ones under bending loads.

B. The type of loads that can be applied over the column are the following ones:

Uniform Column:

a) Axial loads:

- Tension or compression load at the ends of the column

b) Transversal loads:

- Discrete loads, applied at any point along the length of the column length
- Distributed loads along the whole length of part of it.
- Discrete bending moments, applied at any point along the column length

Non Uniform Column:

a) Axial loads:

- Cargas concentradas, aplicadas en cualquier punto de la longitud de la columna.
- Cargas distribuidas sobre toda o parte de la longitud de la columna.

b) Transversal loads: Los mismos tipos que en columnas uniformes.

C. The type of analysis performed by the program are:

1. Column buckling: The program calculates the Reserve Factor as a column buckling, taking into account an elastic-plastic behaviour of the material. Column buckling loads will be the applied ones multiplied by the reserve factor obtained. Column buckling modes are:

Uniform Column:

- a. Bending buckling respect to Y and Z axis
- b. Torsion buckling
- c. Interaction between bending and crippling
- d. Buckling under bending and torsions loads

Non Uniform Column:

- a. Bending buckling respect to Y axis

2. Beam-Column strength: The programs calculates the internal loads, stresses and deflexion of the beam-column resultant of the axial and transversal applied loads, assuming a behaviour completely elastic of the column.

If there are no axial loads applied, the program analyses the beam under transversal loads only.

The programs uses the successive approximation method of Newmark to calculate the bending reserve factor and to perform the beam-column analysis. This method follows an iterative process that ends when the relative variations obtained in the last two iterations is lower than $0.5E-4$ (result is correct till the fourth significant digit). The compared values are the reserve factor in column buckling and the mean deflexion in the strength analysis.

Differences between versions:

- **Version 3.0:**

Now, the user can choose the language for screen dialogs and for the output file, between Spanish and English.

The data flow of the program is modified. Now, the input data as well as the output results are provided by means of files:

- Input file: With the input data grouped in the same way as for other applications.
- Output file: With all the input data and output results in printable form.

Since the format and arrangement of the input file is completely different from previous version, input files are not compatible between versions.

The new program source code is structured to allow the creation of a Dynamic Link Library DLL for Windows operating system.

In parallel with this, a Windows application has been developed as Graphic User Interface GUI for the program PANCO. This application uses the same input file and provides the same output results. The analysis is performed by calling to the DLL, compiled from the same source as the official Fortran version.

3.- Program execution

Prior to the program execution, it is necessary to create the input data file, as described in section 4. The program can be then executed by typing its **full path name**.

If the software symbol table has been previously assigned (consult with the System Manager in order to know how to assign this symbol table); it can be executed by typing the command:

```
$panco
```

The program asks for the language to use, the name of the input file and the name of the output file.

```
Idioma / Language:
=====

1 ... ESPANOL
2 ... ENGLISH
Idioma / Language: 2
Name of the input file: Example.dat
Name of the output file: Results.txt
```

Details of the input and output files are provided in next sections. The CPU time for calculation is negligible.

It is possible to execute the program with a single command, including the input and output filenames and the language as parameters. The above example can be executed as follows:

```
$panco Example.dat Results.txt 2
```

If the third parameter (language) is omitted, the Spanish language will be taken by the program.

4.- Input data file

In this chapter, it is described the structure of the input file, which is required for the correct execution of the PANCO program.

This file contains all the necessary data that defines the problem. It basically contains eleven data groups, all of them are necessary. They will be numbered from 1 to 11 and must be in order in the input file.

All the numerical data contained in this file will be read in a free format. The variables will be separated by one or more blank spaces. The numbering of each group will be included in the first column.

Note that groups 5, 6 and 7 depends on the value assumed in Group 2.

Listed below are the contents of each data group:

- **GROUP 1: Title of the problem**

Four lines, containing the title of the problem to be solved, which will appear at the header of the output file:

Line 1 1 First line of the title

Line 2 Second line of the title

Line 3 Third line of the title

Line 4 Fourth line of the title

- **GROUP 2: Type of the Column**

Line 1 2 TYPE OF COLUMN: 1 Uniform, 2 Non Uniform

Line 2 1 (if the column is Uniform) or 2 (if the column is Non Uniform)

- **GROUP 3: Units**

Line 1 3 UNITS (N/daN/Kgf/lbf/klb, mm/cm/in)

Line 2 Force Units and Length Units (i.e. N,mm)

- **GROUP 4: Material Data**

Line 1 4 MATERIAL DATA

Line 2 Et Ec G Ftu Fty Fcy e

• **GROUP 5: Geometry**

Line 1 5 GEOMETRY

Then, if Type of Column = 1: Uniform column

Line 2 L A I_{yy} I_{zz} J R y_0 z_0 Fcc Ds Di

The values have to comply with the following requirements:

$L > 0$

$A > 0$

$I_{yy} \geq 0$ If it is equal to 0, the following analysis will not be performed:

- Bending buckling about Y Axis
- Torsion buckling
- Bending + Torsion buckling
- Buckling about Y axis + Crippling
- Beam-Column Bending about Y axis

$I_{zz} \geq 0$ If it is equal to 0, the following analysis will not be performed:

- Bending buckling about Z Axis
- Torsion buckling
- Bending + Torsion buckling
- Buckling about Z axis + Crippling
- Beam-Column Bending about Z axis

$J \geq 0$ If $J = 0$ and $R = 0$, the following analysis will not be performed:

- Torsion buckling
- Bending + Torsion buckling

$R \geq 0$ If $J = 0$ and $R = 0$, the following analysis will not be performed:

- Torsion buckling
- Bending + Torsion buckling

If $y_0 = 0$, there is no interaction between the buckling about Y axis and Torsion buckling

If $z_0 = 0$, there is no interaction between the buckling about Z axis and Torsion buckling

If $F_{cc} = 0$, the following analysis will not be performed:

- Bending + Crippling buckling

If $D_s = 0$ or $D_i = 0$ the stresses due to the beam-column bending will be zero at the corresponding fiber.

Then, if Type of Column = 2: Non Uniform column

Line 2	N_g	x_1	A_1	I_1	DS_1	DI_1
Line i		x_i	A_i	I_i	DS_i	DI_i
Line Ng		x_{Ng}	A_{Ng}	I_{Ng}	DS_{Ng}	DI_{Ng}

The values have to comply with the following requirements:

$$X \geq 0$$

$$A > 0$$

$$I \geq 0 \quad \text{If } I = 0 \text{ the following analysis will not be performed:}$$

- Bending buckling about Y axis

- Beam-Column bending about Y axis

If $DS_i = 0$ or $DI_i = 0$, the stresses due to the beam-column bending will be zero at the corresponding fiber.

• **GROUP 6:**

Line 1 6 WORK PLANE AND SUPPORT CONDITIONS:

Then, if Type of Column = 1: Uniform column

Line 2 WP (WP = 1 for XZ working plane and WP = 2 for XY working plane)

Line 3 SPy, SPz (support conditions about Y axis Bending and Z axis Bending)

SPi = 1 Simply supported - Simply supported

SPi = 2 Clamped - Clamped

SPi = 3 Simply supported - Clamped

SPi = 4 Free - Clamped

SPi = 5 Simply supported - Supported

SPi = 6 Supported - Clamped

SPi = 7 Supported - Supported

Line 4 SPx (support conditions about X axis Torsion)

SPx = 1 Free warping - Free warping

SPx = 2 Constrained warping - Constrained warping

SPx = 3 Free warping - Constrained warping

SPx = 4 Constrained warping - Free warping

Then, if Type of Column = 2: Non Uniform column

Line 2 SPy (support conditions about Y axis Bending and Z axis Bending, as above)

- **GROUP 7:**

Line 1 7 DISCRETE AXIAL LOADS

Then, if Type of Column = 1: Uniform column

Line 2 1 x_{Px1} F_{x1}

F_{x1} is negative for compression loads and positive for tension loads.

Then, if Type of Column = 2: Non Uniform column

Line 2 N_{Px} x_{Px1} F_{x1}

Line i x_{Pxi} F_{xi}

Line N_{Px} $x_{Px,NPx}$ $F_{x,NPx}$

Positive loads have positive sign of X axis. Data will be typed in ascending order of x coordinate.

- **GROUP 8:**

If the Type of Column = 1, then, this Group **must not appear**.

Line 1 8 DISTRIBUTED AXIAL LOADS

Line 2 N_{qx} x_{qx1} q_{x1}

Line i x_{qxi} q_{xi}

Line N_{qx} $x_{qx,Nqx}$ $q_{x,Nqx}$

Positive loads have positive sign of X axis. Data will be typed in ascending order of x coordinate. It will be considered that the load varies linearly along the column length.

Apart from the initial and final sections, the data for Sections in which any change is produced must be written also, and the total length of the column must be represented.

- **GROUP 9:**

Line 1 9 DISCRETE TRANSVERSAL LOADS

Line 2 N_{Pt} x_{Pt1} P_{t1}

Line i x_{Pti} P_{ti}

Line N_{Pt} $x_{Pt,NPt}$ $P_{t,NPt}$

Positive loads have negative sign of Y (or Z) axis. Data will be typed in ascending order of x coordinate.

- **GROUP 10:**

Line 1 10 DISTRIBUTED TRANSVERSAL LOADS

Line 2 N_{qt} x_{qt1} q_{t1}

Line i x_{qti} q_{ti}

Line N_{qt} $x_{qt,Nqt}$ $q_{x,Nqt}$

Positive loads have negative sign of Y (or Z) axis. Data will be typed in ascending order of x coordinate.

Apart from the initial and final sections, the data for Sections in which any change is produced must be written also, and the total length of the column must be represented.

- **GROUP 11:**

Line 1 11 DISCRETE BENDING MOMENTS

Line 2 N_M x_{M1} P_{M1}

Line i x_{Mi} P_{Mi}

Line N_M $x_{M,NM}$ $P_{M,NM}$

Positive moments have negative (positive) sign of Y (Z) axis for a working plane X. Data will be typed in ascending order of x coordinate.

5.- Output results

This section includes the description of the output result file as well as the possible error Messages.

5.1 Output file

The PANCO program creates one printable file with all the results of the analysis carried out. This file includes the following information:

- **Summary of the input data:**

- Type of column
- Material data
- Geometric Data
- Support conditions
- Applied loads & moments

- **Column Buckling analysis:**

Uniform Column:

- Buckling under Bending about Y axis Reserve Factor
- Buckling under Bending about Z axis Reserve Factor
- Buckling under Torsion about X axis Reserve Factor
- Buckling under Bending + Crippling about Y axis Reserve Factor
- Buckling under Bending + Crippling about Z axis Reserve Factor
- Buckling under Torsion + Bending Reserve Factor

Non Uniform Column:

- Buckling under Bending about Y axis Reserve Factor

For each RF, the program provides as well the corresponding values of the stress and the resultant elastic modulus corrected by plasticity (if it's required).

- **Results:**

Along the length of the column, for the significant X coordinates, the program provides:

- Deflexion
- Axial force
- Bending moment
- Shear force
- Stress at Upper fiber
- Stress at Lower fiber

5.2 Error messages

When the program detects an error in the input data, it writes the corresponding message in the console and stops the execution. The error messages that might appear are listed below:

Error messages:

```

ERROR: The input file cannot be opened
ERROR: Erroneous group number in the input file
ERROR: No Group 1 in the input file
ERROR: End of file while reading Group 1 in the input file
ERROR: No Group 2 in the input file
ERROR: End of file while reading Group 2 in the input file
ERROR: Bad definition of Group 2
ERROR: No Group 3 in the input file
ERROR: End of file while reading Group 3 in the input file
ERROR: Bad definition of Group 3
ERROR: No Group 4 in the input file
ERROR: End of file while reading Group 4 in the input file
ERROR: Bad definition of Group 4
ERROR: No Group 5 in the input file
ERROR: End of file while reading Group 5 in the input file
ERROR: Bad definition of Group 5
ERROR: Group 5: Number of Sections higher than 100
ERROR: Group 5: Erroneous column length
ERROR: No Group 6 in the input file
ERROR: End of file while reading Group 6 in the input file
ERROR: Bad definition of Group 6
ERROR: Grupo 6: Bad definition of working plane
ERROR: Group 6: Support in Y axis bad defined
ERROR: Group 6: Support in Z axis bad defined
ERROR: Group 6: Support in Torsion bad defined
ERROR: End of file while reading group 7 in the input file
ERROR: Bad definition of Group 7
ERROR: Group 7: Number of Sections higher than 100
ERROR: Group 7: Erroneous number of Sections
ERROR: End of file while reading group 8 in the input file
ERROR: Bad definition of Group 8
ERROR: Group 8: Number of Sections higher than 100
ERROR: Group 8: Erroneous number of Sections
ERROR: End of file while reading group 9 in the input file
ERROR: Bad definition of Group 9
ERROR: Group 9: Number of Sections higher than 100
    
```

ERROR: Group 9: Erroneous number of Sections
ERROR: End of file while reading group 10 in the input file
ERROR: Bad definition of Group 10
ERROR: Group 10: Number of Sections higher than 100
ERROR: Group 10: Erroneous number of Sections
ERROR: End of file while reading group 11 in the input file
ERROR: Bad definition of Group 11
ERROR: Group 11: Number of Sections higher than 100
ERROR: Group 11: Erroneous number of Sections
ERROR: No applied loads
ERROR: Undefined working plane
ERROR: Undefined support conditions
ERROR: Et = 0
ERROR: Ec = 0
ERROR: G = 0
ERROR: Ftu = 0
ERROR: Fty = 0
ERROR: Fcy = 0
ERROR: e = 0
ERROR: Undefined type of analysis
ERROR: Erroneous geometric data
ERROR: There are no axial loads applied
ERROR: Iyy = 0 o Izz = 0
ERROR: Maximum deflexion is null
ERROR: Area calculation: Low Limit > Up Lim
ERROR: Area calculation: Disordered abscissas
ERROR: Area calculation: Low Limit out of interval
ERROR: Area calculation: Up Limit out of interval
ERROR: Area calculation: Different sub-intervals
ERROR: Convergence failure obtaining the root
ERROR: Bending-Torsion buckling: Imaginary root
ERROR: Bending-Torsion buckling: Null root
ERROR: Null Maximum stress
ERROR: It's not possible to open the output file
ERROR: The abscissas are not sorted
ERROR: Abscissa to be interpolated is out of interval'

6.- Windows application

An application has been developed in Visual Basic .NET for Windows environment. The original routines of the program for calculation and output results have been included in a Dynamic Link Library, which is used by this application. Therefore the same results will be obtained, with only differences due to rounding.

Since this is a Windows application, it must be installed in the user's PC before it can be executed. For installation, please contact with the system manager.

The application can be executed by "double click" in the file **Panco.exe**, located in the PC's server, or by means a **link** in the "desktop" or in the "start menu".

The input data must be typed in the application form or retrieved from an input file. The form, after loading the data corresponding to the example 1 (see section 7), has the following appearance:

Panco - Ejemplo 1en.pnc

Titulo Example 1
 Uniform Column
 Airbus Military
 May 2011

Idioma
☒ Español
☐ English

Material
 Et 71710 G 27580 Fty ... 400.0 e (%) .. 7.00
 Ec 73770 Ftu ... 476.0 Fcy .. 400.0

Unidades
 Fuerza N
 Longitud mm

☒ Seccion Uniforme ☐ Seccion No Uniforme

Geometria - Seccion Uniforme:
 L 500 Y0 ... -11.698
 A 187.5 Z0 ... 0
 Iyy ... 44140 Fcc ... 363
 Izz ... 6951 Ds ... 20
 J 390.6 Di ... 20
 R 1689790

Apoyos: Flexion-Y
 Simplem apoyado - Simplem apoyado

Apoyos: Flexion-Z
 Simplem apoyado - Simplem apoyado

Apoyos: Torsion
 Alab. permitido - Alab. permitido

Cargas

Tipo	X	F
Fuerza axial puntual	0	-15000
Momento puntual: My	0	-100000
Momento puntual: My	500	200000

Resultados

Pandeo de Columna **Resistencia de Viga-Columna**

	S	Et	RF
RF-Fly	-358.8	38605.8	4.48
RF-Flz	-108.0	73770.0	1.35
RF-T	-204.5	73768.8	2.56
RF-FlyCC	-315.0	68794.9	3.94
RF-FlzCC	NA	NA	NA
RF-FIT	-182.4	73769.9	2.28

Figure 2: Input data form.

The program calculates the results for all the included loads at the moment it has all the required data. In example 1, there are one axial load and a couple of discrete bending moments and the user can read the results at the botom framebox.

7.- Examples

Input data and obtained results are presented for the two following examples.

7.1 Example 1

In the first example it will be analyzed a Uniform Column, simply supported at both ends. The applied loads are showed in Figure 3 below:

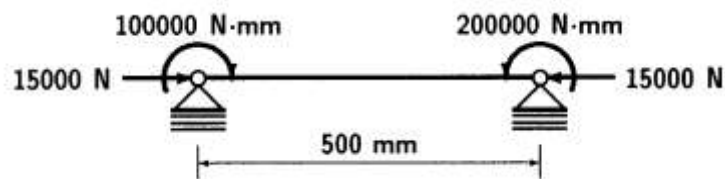


Figure 3: Example 1

The material properties of the column are:

$E_t =$	71710	N/mm^2
$E_c =$	73770	N/mm^2
$G =$	27580	N/mm^2
$F_{tu} =$	476	N/mm^2
$F_{ty} =$	400	N/mm^2
$F_{cy} =$	400	N/mm^2
$e =$	7 %	

The length of the column is $L = 500 \text{ mm}$, and the cross section geometric properties are:

$A =$	187.5	mm^4
$I_{yy} =$	44140.0	mm^4
$I_{zz} =$	6951.0	mm^4
$J =$	390.6	mm^4
$R =$	1689790.0	mm^4
$Y_o =$	-11.698	mm
$Z_o =$	0.0	mm
$F_{cc} =$	363.0	N/mm^2
$D_s =$	20.0	mm
$D_{j.} =$	20.0	mm

Input file:

```

1 Example 1
  Uniform Column
  Airbus Military
  May 2011
2 TYPE OF COLUMN: 1 Uniform, 2 Non Uniform
  1
3 UNITS (N/daN/Kgf/lbf/klb,mm/cm/in)
  N , mm
4 MATERIAL DATA
  71710, 73770, 27580, 476.0, 400.0, 400.0, 7.00
5 GEOMETRY
  500.0, 187.5, 44140, 6951, 390.6, 1689790, -11.698, 0.0, 363.0, 20.0, 20.0
6 WORK PLANE AND SUPPORT CONDITIONS: 1-XZ/2-YZ ; Bend-Y, Bend-Z / Tors-X
  1
  1, 1
  1
7 DISCRETE AXIAL LOADS
  1,      0.00,      -15000.0
11 DISCRETE BENDING MOMENTS
  2,      0.00,      -100000.00
      500.00,      200000.00

```

Output file:

```

+-----+
|               |
|   PROGRAM PANCO   |
|               |
|   VERSION  3.0    |
|               |
|   MAY-2011       |
|               |
+-----+

```

DATA ENTRY

=====

Type of column:

Uniform (Constant Section and discrete axial load at both ends)

Column Material Properties:

Tension Elastic Modulus:	Et =	71710.0	N/mm ²
Compression Elastic Modulus:	Ec =	73770.0	N/mm ²
Shear Elastic Modulus:	G =	27580.0	N/mm ²
Ultimate Stress:	Ftu =	476.00	N/mm ²
Tension Yield Stress:	Fty =	400.00	N/mm ²
Compression Yield Stress:	Fcy =	400.00	N/mm ²
Permanent Strain at Ultimate Stress:	e =	7.00	%

Column Geometry Data:

Column Length:	L =	500.00	mm
Area of Cross Section:	A =	187.50	mm ²
M. of Inertia about Y-axis:	Iyy =	44140.00	mm ² /mm ²
M. of Inertia about Z-axis:	Izz =	6951.00	mm ² /mm ²
Torsion Constant:	J =	390.60	mm ² /mm ²
Warping Constant:	R =	1689790.00	mm ³ /mm ³
Y coordinate from Shear Center to c.o.g.:	Yo =	-11.70	mm
Z coordinate from Shear Center to c.o.g.:	Zo =	0.00	mm
Crippling Stress of the column:	Fcc =	363.00	N/mm ²
Distance from Neutral Axis to Upper Fiber:	Ds =	20.00	mm
Distance from Neutral Axis to Lower Fiber:	Di =	20.00	mm

Support conditions at both ends of the column:

	Xs = 0	Xs = L
Flex. resp. al eje Y:	Simply supported	Simply supported
Flex. resp. al eje Z:	Simply supported	Simply supported
Torsion:	Free warping	Free warping
	Gyration is not allowed at both ends always	

Applied loads over the column:

XS (mm)	AXIAL LOADS		TRANSVERSAL LOADS		
	Px (N)	px (N/mm)	Pt (N)	pt (N/mm)	Mf (N·mm)
0.000	15000.00	0.00	0.00	0.00	-100000.00
500.000	-15000.00	0.00	0.00	0.00	200000.00

COLUMN BUCKLING ANALYSIS

=====

Stresses, Tangent Modulus and Reserve Factors for buckling loads

BUCKLING MODE	S (N/mm ²)	Et (N/mm ²)	R.F.
Bending about Y - axis	-358.8	38605.8	4.48
Bending about Z - axis	-108.0	73770.0	1.35
Torsion	-204.5	73768.8	2.56
Bending-Y + Crippling	-315.0	68794.9	3.94
Bending-Z + Crippling	---	---	---
Bending-Y + Torsion	-182.4	73769.9	2.28

BEAM-COLUMN STRENGTH ANALYSIS
=====

Maximum and Minimum summary

		MAXIMUM		MINIMUM		
		XS	VALUE	XS	VALUE	
Deflexion	(DELTA)	0.000	0.00	262.500	-1.67	mm
Axial Force	(FA)	0.000	-15000.00	0.000	-15000.00	N
Bending Moment	(MF)	487.500	200106.42	0.000	100000.00	N·mm
Shear Force	(FC)	0.000	374.73	500.000	-14.36	N
Stress Up Fiber	(S_FS)	0.000	-125.31	487.500	-170.67	N/mm ²
Stress Lw Fiber	(S_FI)	487.500	10.67	0.000	-34.69	N/mm ²

Deflexion, Forces, Bending Moments and Stresses for the applied loads

XS (mm)	DELTA (mm)	FA (N)	MF (N·mm)	FC (N)	S_FS (N/mm ²)	S_FI (N/mm ²)
0.000	0.00000	-15000.00	100000.00	374.73	-125.31	-34.69
12.500	-0.14307	-15000.00	104645.97	368.69	-127.42	-32.58
25.000	-0.28104	-15000.00	109215.55	362.51	-129.49	-30.51
37.500	-0.41370	-15000.00	113705.41	356.00	-131.52	-28.48
50.000	-0.54083	-15000.00	118112.27	349.23	-133.52	-26.48
62.500	-0.66221	-15000.00	122432.93	342.20	-135.47	-24.53
75.000	-0.77764	-15000.00	126664.22	334.93	-137.39	-22.61
87.500	-0.88689	-15000.00	130803.05	327.41	-139.27	-20.73
100.000	-0.98979	-15000.00	134846.41	319.65	-141.10	-18.90
112.500	-1.08612	-15000.00	138791.36	311.66	-142.89	-17.11
125.000	-1.17570	-15000.00	142634.98	303.44	-144.63	-15.37
137.500	-1.25834	-15000.00	146374.50	295.00	-146.32	-13.68
150.000	-1.33385	-15000.00	150007.19	286.34	-147.97	-12.03
162.500	-1.40206	-15000.00	153530.39	277.48	-149.57	-10.43
175.000	-1.46281	-15000.00	156941.53	268.41	-151.11	-8.89
187.500	-1.51591	-15000.00	160238.11	259.14	-152.60	-7.40
200.000	-1.56122	-15000.00	163417.73	249.69	-154.05	-5.95
212.500	-1.59858	-15000.00	166478.09	240.06	-155.43	-4.57
225.000	-1.62784	-15000.00	169416.92	230.24	-156.76	-3.24
237.500	-1.64885	-15000.00	172232.11	220.27	-158.04	-1.96
250.000	-1.66148	-15000.00	174921.56	210.13	-159.26	-0.74
262.500	-1.66560	-15000.00	177483.38	199.84	-160.42	0.42
275.000	-1.66108	-15000.00	179915.61	189.39	-161.52	1.52
287.500	-1.64781	-15000.00	182216.53	178.82	-162.56	2.56
300.000	-1.62567	-15000.00	184384.45	168.11	-163.55	3.55
312.500	-1.59455	-15000.00	186417.81	157.28	-164.47	4.47
325.000	-1.55437	-15000.00	188315.08	146.34	-165.33	5.33
337.500	-1.50502	-15000.00	190074.89	135.28	-166.12	6.12
350.000	-1.44643	-15000.00	191695.97	124.14	-166.86	6.86
362.500	-1.37850	-15000.00	193177.14	112.89	-167.53	7.53
375.000	-1.30117	-15000.00	194517.28	101.57	-168.14	8.14
387.500	-1.21438	-15000.00	195715.44	90.17	-168.68	8.68
400.000	-1.11807	-15000.00	196770.75	78.71	-169.16	9.16
412.500	-1.01218	-15000.00	197682.44	67.19	-169.57	9.57
425.000	-0.89667	-15000.00	198449.81	55.62	-169.92	9.92
437.500	-0.77150	-15000.00	199072.34	44.01	-170.20	10.20
450.000	-0.63665	-15000.00	199549.58	32.36	-170.42	10.42
462.500	-0.49208	-15000.00	199881.14	20.70	-170.57	10.57
475.000	-0.33779	-15000.00	200066.81	9.01	-170.65	10.65
487.500	-0.17377	-15000.00	200106.42	-2.67	-170.67	10.67
500.000	0.00000	-15000.00	200000.00	-14.36	-170.62	10.62

7.2 Example 2

In the second Example, it will analyzed a Non Uniform column, fixed at both ends. The applied loads are showed in Figure 3 below:

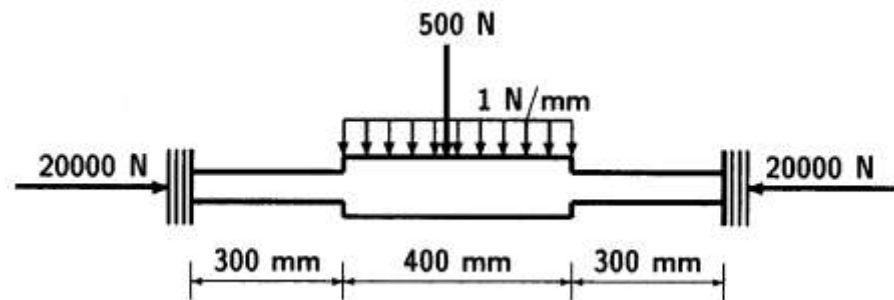


Figure 4: Example 2

The material properties of the column are:

$E_t =$	71710	N/mm:
$E_c =$	73770	N/mm ²
$G =$	27580	N/mm ²
$F_{tu} =$	476	N/mm ²
$F_{ty} =$	400	N/mm ²
$F_{cy} =$	400	N/mm ²
$e =$	7	%

The length of the column is $L = 1000$ mm, and the cross section geometric properties are:

- Sección normal:

$A =$	250	mm ²
$I_{yy} =$	15000	mm ⁴
$DS =$	25	mm
$DI =$	25	mm

- Sección reforzada

$A =$	400	mm ²
$I_{yy} =$	25000	mm ⁴
$DS =$	40	mm
$DI =$	40	mm

Input file:

```

1 Example2
  Non Uniform Column
  Airbus Military
  May 2011
2 TYPE OF COLUMN: 1 Uniform, 2 Non Uniform
  2
3 UNITS (N/daN/Kgf/lbf/klb,mm/cm/in)
  N, mm
4 MATERIAL DATA
  71710, 73770, 27580, 476.0, 400.0, 400.0, 7.00
5 GEOMETRY
  6,
    0.00,      250.0,      15000.0,      25.00,      25.00
    300.00,    250.0,      15000.0,      25.00,      25.00
    300.00,    400.0,      25000.0,      40.00,      40.00
    700.00,    400.0,      25000.0,      40.00,      40.00
    700.00,    250.0,      15000.0,      25.00,      25.00
    1000.00,   250.0,      15000.0,      25.00,      25.00
6 WORK PLANE AND SUPPORT CONDITIONS: 1-XZ/2-YZ ; Bend-Y, Bend-Z / Tors-X
  2
7 DISCRETE AXIAL LOADS
  2,
    0.00,      20000.0
    1000.00,   -20000.0
9 DISCRETE TRANSVERSAL LOADS
  1,
    500.00,      500.0
10 DISTRIBUTED TRANSVERSAL LOADS
  6,
    0.00,      0.0
    300.00,    0.0
    300.00,    1.0
    700.00,    1.0
    700.00,    0.0
    1000.00,   0.0

```

Output file:

```

+-----+
|               |
|  PROGRAM  PANCO  |
|               |
|  VERSION   3.0   |
|               |
|  MAY-2011   |
|               |
+-----+

```

DATA ENTRY

```
=====
```

Type of column:

Uniform (Constant Section and discrete axial load at both ends)

Column Material Properties:

```
-----
```

Tension Elastic Modulus:	Et =	71710.0	N/mm ²
Compression Elastic Modulus:	Ec =	73770.0	N/mm ²
Shear Elastic Modulus:	G =	27580.0	N/mm ²
Ultimate Stress:	Ftu =	476.00	N/mm ²
Tension Yield Stress:	Fty =	400.00	N/mm ²
Compression Yield Stress:	Fcy =	400.00	N/mm ²
Permanent Strain at Ultimate Stress:	e =	7.00	%

Column Geometry Data:

```
-----
```

XS (mm)	AREA (mm ²)	INERTIA (mm ² · mm ²)	DISTANCIA DE EJE NEUTRO A	
			UPPER (mm)	LOWER (mm)
0.00000	250.00	15000.00	25.000	25.000
299.99960	250.00	15000.00	25.000	25.000
300.00000	400.00	25000.00	40.000	40.000
699.99957	400.00	25000.00	40.000	40.000
700.00000	250.00	15000.00	25.000	25.000
1000.00000	250.00	15000.00	25.000	25.000

Support conditions at both ends of the column:

```
-----
```

Xs = 0

Xs = L

Flex. resp. al eje Y:

Clamped

Clamped

Gyratation is not allowed at both ends always

Applied loads over the column:

```
-----
```

XS (mm)	AXIAL LOADS		TRANSVERSAL LOADS		
	Px (N)	px (N/mm)	Pt (N)	pt (N/mm)	Mf (N · mm)
0.000	20000.00	0.00	0.00	0.00	0.00
300.000	0.00	0.00	0.00	0.00	0.00
300.000	0.00	0.00	0.00	1.00	0.00
500.000	0.00	0.00	500.00	1.00	0.00
700.000	0.00	0.00	0.00	1.00	0.00
700.000	0.00	0.00	0.00	0.00	0.00
1000.000	-20000.00	0.00	0.00	0.00	0.00

COLUMN BUCKLING ANALYSIS

=====

BUCKLING MODE

R.F.

Bending about Y-axis

2.83

Stresses, Tangent Modulus and Reserve Factors for buckling loads

XS (mm)	S (N/mm ²)	Et (N/mm ²)
0.000	-226.05	73761.5
25.000	-226.05	73761.5
50.000	-226.05	73761.5
75.000	-226.05	73761.5
100.000	-226.05	73761.5
125.000	-226.05	73761.5
150.000	-226.05	73761.5
175.000	-226.05	73761.5
200.000	-226.05	73761.5
225.000	-226.05	73761.5
250.000	-226.05	73761.5
275.000	-226.05	73761.5
300.000	-226.05	73761.5
300.000	-141.28	73770.0
325.000	-141.28	73770.0
350.000	-141.28	73770.0
375.000	-141.28	73770.0
400.000	-141.28	73770.0
425.000	-141.28	73770.0
450.000	-141.28	73770.0
475.000	-141.28	73770.0
500.000	-141.28	73770.0
525.000	-141.28	73770.0
550.000	-141.28	73770.0
575.000	-141.28	73770.0
600.000	-141.28	73770.0
625.000	-141.28	73770.0
650.000	-141.28	73770.0
675.000	-141.28	73770.0
700.000	-141.28	73770.0
700.000	-226.05	73761.5
725.000	-226.05	73761.5
750.000	-226.05	73761.5
775.000	-226.05	73761.5
800.000	-226.05	73761.5
825.000	-226.05	73761.5
850.000	-226.05	73761.5
875.000	-226.05	73761.5
900.000	-226.05	73761.5
924.999	-226.05	73761.5
949.998	-226.05	73761.5
974.997	-226.05	73761.5
1000.000	-226.05	73761.5

BEAM-COLUMN STRENGTH ANALYSIS
=====

Maximum and Minimum summary

		MAXIMUM		MINIMUM		
		XS	VALUE	XS	VALUE	
Deflexion	(DELTA)	999.998	0.00	500.000	-4.87	mm
Axial Force	(FA)	0.000	-20000.00	0.000	-20000.00	N
Bending Moment	(MF)	500.000	160420.12	0.000	-141972.88	N·mm
Shear Force	(FC)	225.000	754.83	775.000	-754.83	N
Stress Up Fiber	(S_FS)	0.000	156.62	500.000	-306.67	N/mm ²
Stress Lw Fiber	(S_FI)	500.000	206.67	0.000	-316.62	N/mm ²

Deflexion, Forces, Bending Moments and Stresses for the applied loads

XS (mm)	DELTA (mm)	FA (N)	MF (N·mm)	FC (N)	S FS (N/mm ²)	S FI (N/mm ²)
0.000	0.00000	-20000.00	-141972.88	451.84	156.62	-316.62
25.000	-0.03951	-20000.00	-129932.63	513.04	136.55	-296.55
50.000	-0.15345	-20000.00	-116403.89	571.07	114.01	-274.01
75.000	-0.33398	-20000.00	-101543.47	621.21	89.24	-249.24
100.000	-0.57260	-20000.00	-85521.14	664.25	62.54	-222.54
125.000	-0.86015	-20000.00	-68520.27	699.74	34.20	-194.20
150.000	-1.18695	-20000.00	-50734.44	727.21	4.56	-164.56
175.000	-1.54278	-20000.00	-32368.17	746.32	-26.05	-133.95
200.000	-1.91711	-20000.00	-13631.75	754.83	-57.28	-102.72
225.000	-2.29925	-20000.00	5260.77	754.83	-88.77	-71.23
250.000	-2.67837	-20000.00	24093.05	752.08	-120.16	-39.84
275.000	-3.04372	-20000.00	42649.75	735.22	-151.08	-8.92
300.000	-3.38465	-20000.00	60718.02	712.96	-181.20	21.20
300.000	-3.38465	-20000.00	60718.29	711.78	-147.15	47.15
325.000	-3.69849	-20000.00	77932.36	666.39	-174.69	74.69
350.000	-3.98554	-20000.00	93985.75	618.86	-200.38	100.38
375.000	-4.24035	-20000.00	108769.23	565.84	-224.03	124.03
400.000	-4.45781	-20000.00	122180.75	508.95	-245.49	145.49
425.000	-4.63334	-20000.00	134128.78	448.53	-264.61	164.61
450.000	-4.76277	-20000.00	144529.84	385.02	-281.25	181.25
475.000	-4.84260	-20000.00	153313.88	318.42	-295.30	195.30
500.000	-4.86979	-20000.00	160420.00	250.69	-306.67	206.67
500.000	-4.86979	-20000.00	160420.12	-250.65	-306.67	206.67
525.000	-4.84263	-20000.00	153314.44	-318.43	-295.30	195.30
550.000	-4.76278	-20000.00	144529.92	-385.05	-281.25	181.25
575.000	-4.63332	-20000.00	134128.34	-448.52	-264.61	164.61
600.000	-4.45781	-20000.00	122180.75	-508.91	-245.49	145.49
625.000	-4.24037	-20000.00	108769.66	-565.84	-224.03	124.03
650.000	-3.98555	-20000.00	93985.89	-618.89	-200.38	100.38
675.000	-3.69848	-20000.00	77932.18	-666.40	-174.69	74.69
700.000	-3.38466	-20000.00	60718.57	-711.75	-147.15	47.15
700.000	-3.38465	-20000.00	60718.32	-712.96	-181.20	21.20
725.000	-3.04373	-20000.00	42650.05	-735.22	-151.08	-8.92
750.000	-2.67838	-20000.00	24093.22	-752.10	-120.16	-39.84
775.000	-2.29924	-20000.00	5260.69	-754.83	-88.77	-71.23
800.000	-1.91711	-20000.00	-13631.71	-754.83	-57.28	-102.72
825.000	-1.54276	-20000.00	-32368.45	-746.33	-26.05	-133.95
850.000	-1.18695	-20000.00	-50734.36	-727.19	4.56	-164.56
875.000	-0.86016	-20000.00	-68519.93	-699.74	34.20	-194.20
900.000	-0.57260	-20000.00	-85521.02	-664.29	62.54	-222.54
924.999	-0.33395	-20000.00	-101543.41	-621.22	89.24	-249.24
949.998	-0.15346	-20000.00	-116402.61	-571.02	114.00	-274.00
974.997	-0.03955	-20000.00	-129930.33	-513.03	136.55	-296.55
1000.000	0.00000	-20000.00	-141972.69	-451.90	156.62	-316.62